Huanxuan Liao

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Education

Institute of Automation, Chinese Academy of Sciences (CASIA) 2023 – Present
PhD student in Pattern Recognition and Intelligent Systems (NLPR). Advisors: Shizhu He and Kang Liu.

North China Electric Power University (NCEPU) 2019 – 2023
Bachelor of Engineering in Intelligence Science and Technology (Innovation Class). Rank: **1/13**.

Research Interests

I work toward foundation models for **long-horizon adaptive intelligence**: models that use growing contexts effectively, retain reusable capabilities, and act reliably over extended interactions.

Long-Context Data & Model Co-Design: data curation and synthesis, length- and capability-aware pretraining and mid-training mixtures, efficient attention and KV caches, and architectures that improve long-context capability, scalability, and efficiency (LCLM-Horizon, HyCo$_2$, SparK, ResAdapt, AllocEmbed).

Knowledge & Continual Adaptation: acquiring, transferring, internalizing and updating knowledge and capabilities, balancing plasticity, retention and generalization (TAGI, SHIP, DyPRAG, AAG, NesyCD, SKIntern, SpaRTA).

Agent Learning & Self-Improvement: persistent memory, long-horizon decisions and learning from interaction and feedback, toward verifiable improvement of agent memory, policies and learning procedures (CITI, KATE, BuPO, Tool-RL-Box, OPERATE, Distill Yourself, SLE).

Selected Publications and Working Papers

Full publication list: personal homepage.

Long-Context Data and Model Co-Design

[13] **[ACM TOIS'26]** Huanxuan Liao, Yao Xu, Shizhu He, Jun Zhao, and Kang Liu. "HyCo$_2$: Hybrid Context Compression with Global Semantics and Local Evidence".

[12] **[arXiv'26]** Huanxuan Liao, Zhongtao Jiang, Yupu Hao, Yuqiao Tan, Shizhu He, Ben Wang, Jun Zhao, Kun Xu, and Kang Liu. "ResAdapt: Adaptive Resolution for Efficient Multimodal Reasoning".

[11] **[arXiv'26]** Song Jin, Zhongtao Jiang, Chenglei Shen, Huanxuan Liao, Haozhe Chi, Zhiwei Wang, Kun Xu, and Yong Liu. "Allocate Before You Embed: Adaptive Visual Input Allocation for Video Embeddings".

[10] **[AAAI'26]** Huanxuan Liao, Yixing Xu, Shizhu He, Guanchen Li, Xuanwu Yin, Dong Li, Emad Barsoum, Jun Zhao, and Kang Liu. "SparK: Query-Aware Unstructured Sparsity with Recoverable KV Cache Channel Pruning".

[9] **[arXiv'25]** Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, Huanxuan Liao, et al. "A Comprehensive Survey on Long Context Language Modeling". *Core contributor to the survey.*
Awesome-LCLM: **Curator & Maintainer**.

Knowledge and Continual Adaptation

[8] **[ACL'26]** Huanxuan Liao, Shizhu He, Yupu Hao, Yequan Wang, Wenhao Teng, Xiangwen Liao, Jun Zhao, and Kang Liu. "Spectral Disentanglement: Rank-Aware Task Adaptation for Rehearsal-free Continual Learning in LLMs".

[7] **[AAAI'25]** Huanxuan Liao, Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, and Jun Zhao. "Neural-Symbolic Collaborative Distillation: Advancing Small Language Models for Complex Reasoning Tasks".

[6] **[COLING'25]** Huanxuan Liao, Shizhu He, Yupu Hao, Xiang Li, Yuanzhe Zhang, Jun Zhao, and Kang Liu. "SKIntern: Internalizing Symbolic Knowledge for Distilling Better CoT Capabilities into Small Language Models".

[5] **[COLING'25]** Huanxuan Liao, Shizhu He, Yao Xu, Yuanzhe Zhang, Shengping Liu, Kang Liu, and Jun Zhao. "Awakening Augmented Generation: Learning to Awaken Internal Knowledge of Large Language Models for Question Answering".

[4] **[NeurIPS'24]** Huanxuan Liao, Shizhu He, Yao Xu, Yuanzhe Zhang, Yanchao Hao, Shengping Liu, Kang Liu, and Jun Zhao. "From Instance Training to Instruction Learning: Task Adapters Generation from Instructions".

Agent Learning and Self-Improvement

[3] **[EMNLP'26]** Yuqiao Tan, Minzheng Wang, Shizhu He, Huanxuan Liao, Chengfeng Zhao, Qiunan Lu, Tian Liang, Jun Zhao, and Kang Liu. "Bottom-up Policy Optimization: Your Language Model Policy Secretly Contains Internal Policies". **Oral.**

[2] **[ACL'26]** Yupu Hao, Zhuoran Jin, Huanxuan Liao, Kang Liu, and Jun Zhao. "Pushing the Limits of LLM Tool Calling via Experiential Knowledge Integration and Activation". *Findings*.

[1] **[ACL'25]** Yupu Hao, Pengfei Cao, Zhuoran Jin, Huanxuan Liao, Yubo Chen, Kang Liu, and Jun Zhao. "Evaluating Personalized Tool-Augmented LLMs from the Perspectives of Personalization and Proactivity".

Selected Awards

- **CAST Young Science and Technology Talent Cultivation Program** (Doctoral Student Special Plan). 2026
- **National Scholarship** (three awards). 2021 / 2022 / 2025
- **Beijing Outstanding Graduate**. Jun. 2023

Research Experience

Tencent (Project UP), Beijing Jul. 2026 – Present

Foundation Model Pretraining and Efficient Inference

Designed multi-QA long-context data and delivered **290B tokens across eight 8K–1M ranges**, with QA structure and mixtures informed by 10B-token ablations. Developed compact block selectors and fused execution; archived MiniCPM/H200 batch-4 tests showed **1.11–1.36× Prefill speedup over Sparse** at 32K–128K. Built an **8.13M-sample SVG corpus (41B tokens)** with SVGFaith-based semantic screening and a **48K-pair sketch-to-code dataset**.

ByteDance, Beijing Jun. 2025 – Jun. 2026

Multimodal Understanding

Deployed efficient multimodal features for AIGC detection and embedding retrieval. Proposed ResAdapt/CAPO to adjust video-frame resolution for efficient reasoning; contributed to AllocEmbed/RDPO for efficient video retrieval. Built OPERATE to evaluate agents in changing environments.

Ant Group, Beijing Feb. 2025 – May 2025

Long-Context Reasoning

Proposed $HyCo_2$ to shorten long inputs while retaining their overall meaning and relevant evidence. Contributed to the LCLM-Horizon survey and curated and maintained Awesome-LCLM.

Professional Services

Conference Reviewer: ACL ARR 2024–2026; NeurIPS 2025–2026; ICML 2026; COLM 2026; ICLR 2025–2026; AAAI 2026.

Open Source: Awesome-LCLM: **Curator & Maintainer**; NVIDIA/kvpress: Contributor; Distill Yourself: Core Logic & Code Contributor.